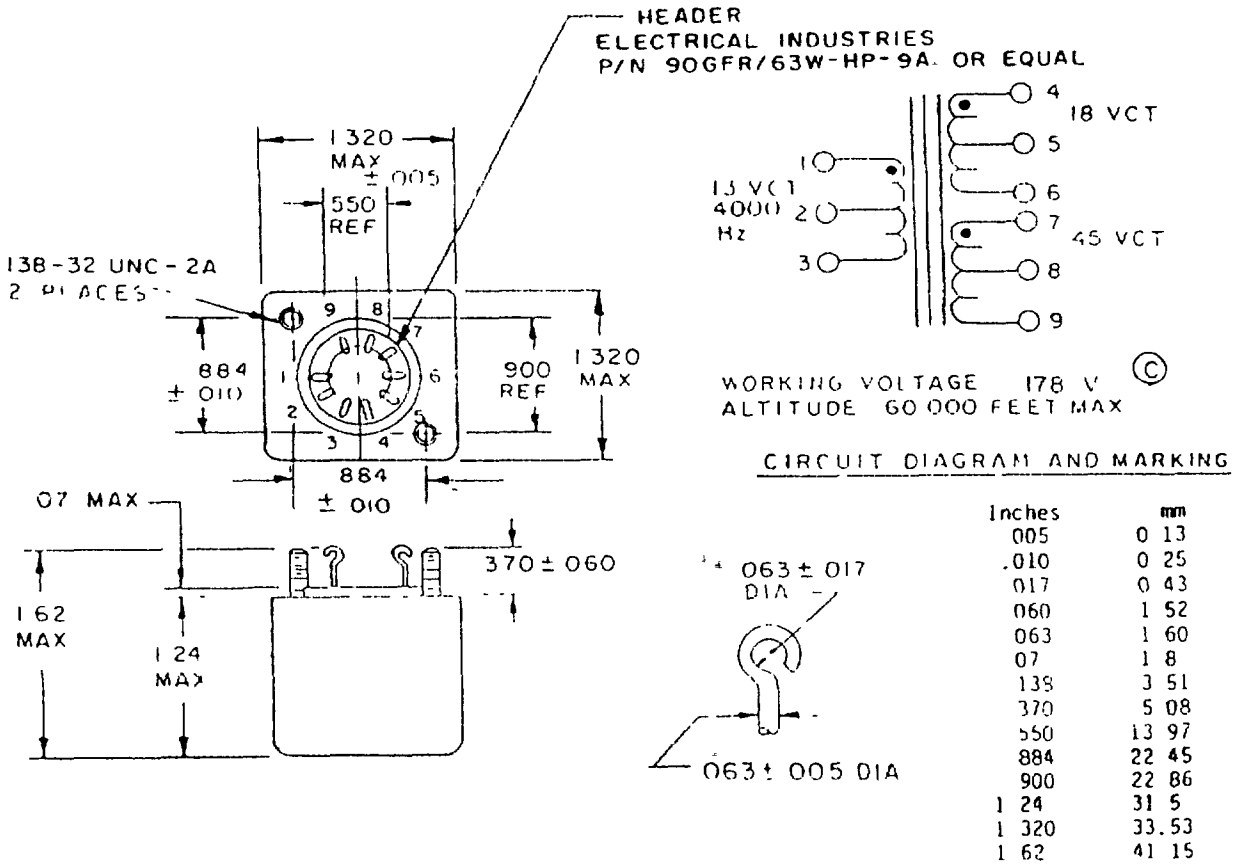
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-UP TF4Y03YY

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-27.



CIRCUIT DIAGRAM AND MARKING

Inches	mm
.005	0.13
.010	0.25
.017	0.43
.060	1.52
.063	1.60
.07	1.8
.138	3.51
.370	5.08
.550	13.97
.884	22.45
.900	22.86
1.24	31.5
1.320	33.53
1.62	41.15

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the sides of the case.

FIGURE 1. Dimensions and configuration

(©) denotes changes

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REQUIREMENTS (When numbers in parentheses, i e , (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings

Primary voltage (1-3) 13 volts rms ct., 4000 Hz

Primary current (1-3). 110 milliamperes maximum.

Secondary voltage

(4-6) 18 volts rms ct.

(7-9) 45 volts rms ct.

Secondary current:

(4-6) 18 ampere

(7-9) 45 ampere.

Working voltage (maximum)

(C) (1-3) 178 volts

(C) (4-6) 178 volts.

(C) (7-9) 178 volts.

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Metal encased.

Terminals Solder terminals, hook type, glass to metal seal, terminals hot tin dipped.

Terminal height .44 inch maximum.

Weight 143 grams maximum.

Operating temperature range -55°C to +155°C

(C) Altitude 60,000 feet maximum.

Terminal strength· MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage

(C) At sea level 500 volts rms

(C) At reduced barometric pressure 300 volts rms.

Electrical characteristics

No load With 13 volts rms ct., 4000 Hz across (1-3), the current in (1-3) shall not exceed 110 milliamperes.

Voltage across

(4-6) 18 volts ct. rms ±2 percent

(7-9) 45 volts ct rms ±2 percent.

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Rated load With 13 volts rms ct., 4000 Hz applied to terminals (1-3), and the current in (1-3) 1 89 amperes, each secondary loaded with 100 ohm ±5 percent resistor, the voltage across

(4 6). 17 85 volts ±? percent

(7-9). 44 volts ±2 percent.

DC resistance

(1-2) = (2-3) .025 ohm maximum.

(4-5) = (5-6) .14 ohm maximum

(7-8) = (8-9) .38 ohm maximum.

Polarity Additive with terminals 3 and 4 connected and terminal 6 connected to terminal /.

Temperature rise 45°C with 13 volts rms, 4000 Hz across (1-3) and the secondaries in a loaded condition at an ambient temperature of 110°C.

Shock (specified pulse) MIL-STD-202, method 213, test condition I.

Moisture resistance Polarizing voltage is not applicable

Marking location See figure 1

Part number M27/311 01

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0714)

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